

A Resonant Counterexample to Closedness of $\pm iA + B$

Solution packet for arXiv:2601.19075

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Status. Candidate counterexample, likely valid; a full negative answer to the closedness uncertainty in Remark 6.3(iii) of the source paper.

1 Source question and setting

Roidos studies, on $Y = L^p(0, T; X)$, the pointwise extension of a closed operator A on X together with

$$Bu = u', \quad \mathcal{D}(B) = W_0^{1,p}(0, T; X),$$

where $u(0) = 0$ [1]. Remark 6.3(iii), on PDF page 29, says that it is not known whether $\pm iA + B$ is closed. The same remark notes that positive definite self-adjoint operators on Hilbert space satisfy the relevant sectorial and R -strip-type hypotheses.

2 Proof intuition

The two unbounded terms can resonate. On an eigenvector with $Ae_n = ne_n$, the phase $e^{-\sigma int}$ makes the large terms in $u' + \sigma iAu$ cancel exactly. We superpose these resonant modes with coefficients $1/n$. The forcing and the solution remain ℓ_2 -valued, but applying A removes the factor $1/n$ and produces the non-square-summable constant sequence. Finite-mode truncations are in the classical intersection domain, so their graph limit proves that the unclosed sum genuinely has a larger closure domain.

3 Counterexample

Theorem 1. *Let $1 < p < \infty$, $T > 0$, $X = \ell_2(\mathbb{N})$, and define*

$$\mathcal{D}(A) = \left\{ x = (x_n) : \sum_{n=1}^{\infty} n^2 |x_n|^2 < \infty \right\}, \quad (Ax)_n = nx_n.$$

Extend A pointwise to $Y = L^p(0, T; X)$, and let $Bu = u'$ with $\mathcal{D}(B) = W_0^{1,p}(0, T; X)$. For each $\sigma \in \{-1, 1\}$, the operator

$$S_\sigma = B + \sigma iA, \quad \mathcal{D}(S_\sigma) = W_0^{1,p}(0, T; X) \cap L^p(0, T; \mathcal{D}(A)),$$

is not closed on Y . Nevertheless, A is positive definite and self-adjoint, $A \in \mathcal{P}(0)$, and $A \in \mathcal{RZ}_c$ for every $c > 0$ in the notation of the source.

Proof. Let $a_n = 1/n$. For $t \in [0, T]$ define

$$u_n(t) = ta_n e^{-\sigma int}, \quad f_n(t) = a_n e^{-\sigma int}.$$

Remark 6.3. (i) Let $c > 0$, X be a complex Banach space, and let $\Lambda \in \mathcal{RV}_c$ in X . Then, by Remark 2.5, we have that $\Lambda^{\frac{1}{2}} \in \mathcal{RZ}_c$. Based on this observation, and on Remark 4.4 (i) and (ii), in Theorem 6.2 we can express all the relevant data in terms of the original operator $A^2 = \Lambda$ instead of A .

(ii) The mixed-derivative regularity that appears in the assumptions of Theorem 6.2 can be obtained by Lemma 4.2 by requiring $\mathcal{BIP}$ for the operator $A^2 = \Lambda$; see Remark 4.4(i).

(iii) Since we do not have a description of the space $E_2^\pm$ in terms of vector-valued Sobolev spaces, and we do not know whether the sum $\pm iA + B$ is closed, the assumptions (i) and (ii) in Theorem 6.2 are natural and necessary in this situation.

(iv) Based on Theorem 5.2 and following the proof of Theorem 6.2, we can establish an existence, uniqueness, and regularity result for a quasilinear version of (6.1)-(6.2). The same holds true for a quasilinear version of (1.1), based on Theorem 5.2 or, even better, on Theorem 5.3. In both cases, we may consider more general forcing terms on the right hand side of the equations; the key point is that certain Fréchet derivatives of the quasilinear operators and of the forcing terms must be locally Lipschitz continuous.

(v) Let X be a Hilbert space and let A be a positive definite self-adjoint linear operator in X . Then $A \in \mathcal{P}(0)$, and the fractional powers defined by (2.2) coincide with the fractional powers defined by the spectral theorem for self-adjoint operators; see, e.g., [1, Theorem III.4.6.7]. In particular, by the spectral theorem for self-adjoint operators, we have that $A \in \mathcal{P}(\theta) \cap \mathcal{BIP}$ for any $\theta \in [0, \pi)$. Recall that, in a Hilbert space, R -boundedness is equivalent to uniform boundedness for a family of bounded operators. Thus, by the spectral theorem for self-adjoint operators, we have that $A \in \mathcal{RZ}_c$ for any $c > 0$. Moreover, the square root $A^{\frac{1}{2}}$ of A is again a positive definite self-adjoint linear operator in X , and hence, $A^{\frac{1}{2}} \in \mathcal{RZ}_c$ for any $c > 0$. Therefore, by Remark 2.5, we obtain that $A \in \mathcal{RV}_c$ for any $c > 0$. $\diamond$

Figure 1: Remark 6.3(iii) states the closedness uncertainty; part (v) records the admissible Hilbert/self-adjoint setting.

Since $a = (a_n) \in \ell_2$, the functions u and f belong to Y ; indeed,

$$\|u(t)\|_{\ell_2} = t\|a\|_{\ell_2}, \quad \|f(t)\|_{\ell_2} = \|a\|_{\ell_2}.$$

Let P_N be the projection onto the first N coordinates and put $u^{(N)} = P_N u$, $f^{(N)} = P_N f$. These finite-coordinate functions belong to $\mathcal{D}(S_\sigma)$ and vanish at $t = 0$. Coordinatewise differentiation gives

$$\frac{d}{dt}(ta_n e^{-\sigma i n t}) + \sigma i n t a_n e^{-\sigma i n t} = a_n e^{-\sigma i n t},$$

and hence

$$S_\sigma u^{(N)} = f^{(N)}.$$

The truncations converge in Y . More explicitly,

$$\|u - u^{(N)}\|_Y = \left(\frac{T^{p+1}}{p+1}\right)^{1/p} \left(\sum_{n>N} \frac{1}{n^2}\right)^{1/2} \longrightarrow 0,$$

while

$$\|f - f^{(N)}\|_Y = T^{1/p} \left(\sum_{n>N} \frac{1}{n^2}\right)^{1/2} \longrightarrow 0.$$

Thus $(u^{(N)}, S_\sigma u^{(N)}) \rightarrow (u, f)$ in $Y \times Y$.

However, for every $t > 0$,

$$(Au(t))_n = nta_n e^{-\sigma int} = te^{-\sigma int},$$

which is not an element of ℓ_2 . Therefore $u \notin L^p(0, T; \mathcal{D}(A))$ and in particular $u \notin \mathcal{D}(S_\sigma)$. The graph of S_σ is not closed.

It remains only to check the source hypotheses. The diagonal operator A is positive definite and self-adjoint. For $\lambda \geq 0$,

$$\|(A + \lambda)^{-1}\| = \sup_{n \geq 1} \frac{1}{n + \lambda} = \frac{1}{1 + \lambda},$$

so $A \in \mathcal{P}(0)$. If $|\operatorname{Im} z| \geq c$, then

$$\|(A + z)^{-1}\| = \sup_{n \geq 1} \frac{1}{|n + z|} \leq \frac{1}{c}.$$

On a Hilbert space uniform boundedness of an operator family implies R -boundedness (apply orthogonality of the Rademacher functions). Hence the resolvent family required in the definition of $\mathcal{RZ}_c$ is R -bounded, and $A \in \mathcal{RZ}_c$ for every $c > 0$. $\square$

4 What the counterexample settles

The answer to the literal closedness uncertainty is negative, and it remains negative for both signs, every $1 < p < \infty$, every $T > 0$, and a positive self-adjoint operator on a Hilbert space. In particular, the use of $\pm iA + B$ in Section 5 of the source is not merely technical under the stated operator assumptions.

The result does not identify $\mathcal{D}(\pm iA + B)$ in Sobolev terms and does not describe the spaces $E_2^\pm$. It also does not claim that the extra embedding assumption in Theorem 6.2 holds for this example.

5 Verification and novelty check

The argument is exact and non-computational. The critical checks are: (1) the sign in the resonant exponential, (2) convergence in the ambient L^p space rather than the intersection graph norm, and (3) failure of the pointwise $\mathcal{D}(A)$ condition for every $t > 0$.

A bounded search on 9 August 2026 used the exact title and arXiv id, the quoted sentence from Remark 6.3(iii), and combinations of `closed sum`, `derivation operator`, `strip-type`, and `$\pm iA + B$` . It found no later paper or exact answer to this May 2026 uncertainty. Novelty is therefore provisional pending expert review of the broader operator-sum literature.

6 Human-review recommendation

Promote after checking that Remark 6.3(iii) is intended under the Section 5 hypotheses exactly as written. The mathematical graph-limit construction is elementary and appears robust; the principal review risk is scope and novelty, not the cancellation calculation.

References

- [1] N. Roidos, *Strip-type operators and abstract Cauchy problems*, arXiv:2601.19075, current version dated 26 May 2026.